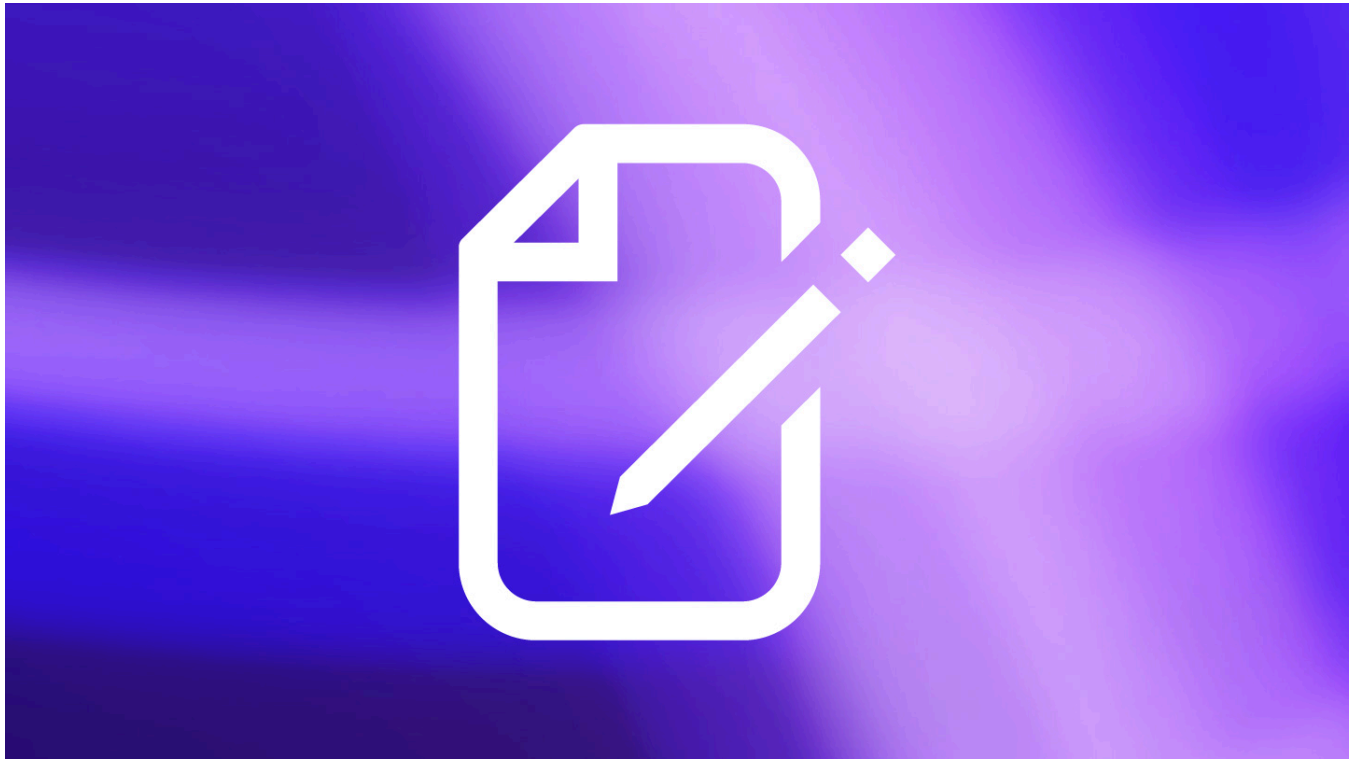
National

Operational Excellence Standard (OES) — Signal configuration

HOW22907 | MODIFIED 11/28/2025



Summary

This standard will cover signal distribution as well as all operational levels required for successful installation of all customer premise equipment.

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What's changed in this article

Date	Summary of change
November 24, 2025	Minor verbiage updates throughout.
September 10, 2025	Added to Celestial.
September 11, 2024	Conducted annual review and updated.
June 23, 2022	Reviewed and made minor updates.
March 19, 2021	Reviewed and made minor updates: • Added guidance for removal of MOCA filter when all IP video is being utilized.
March 5, 2020	Released document.

Overview

This article covers the OES standards for in-premise signal configuration.

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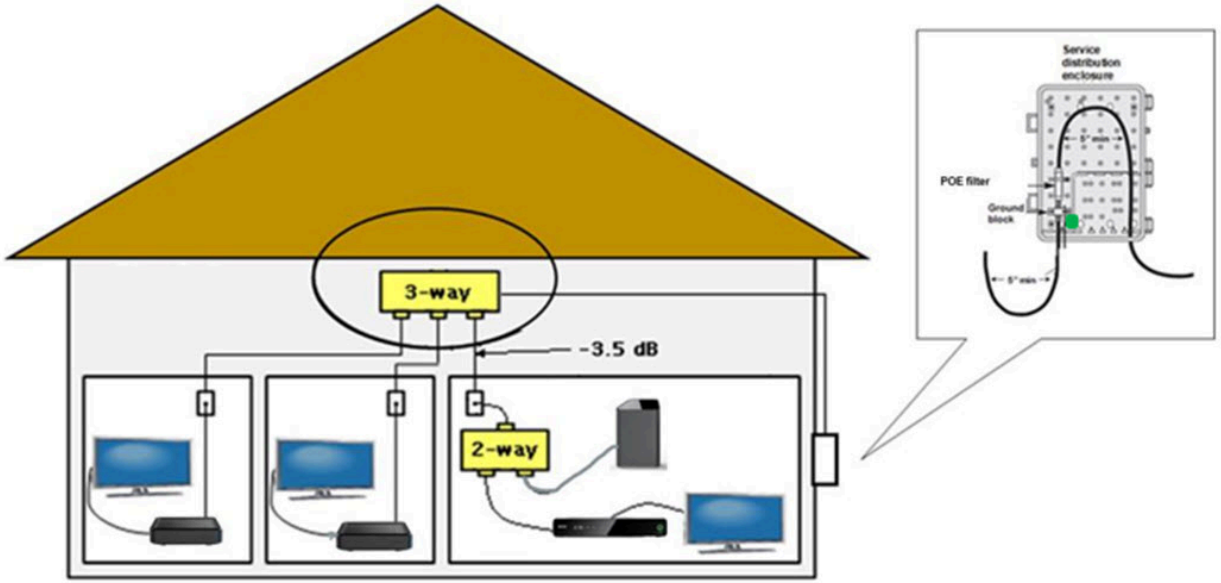
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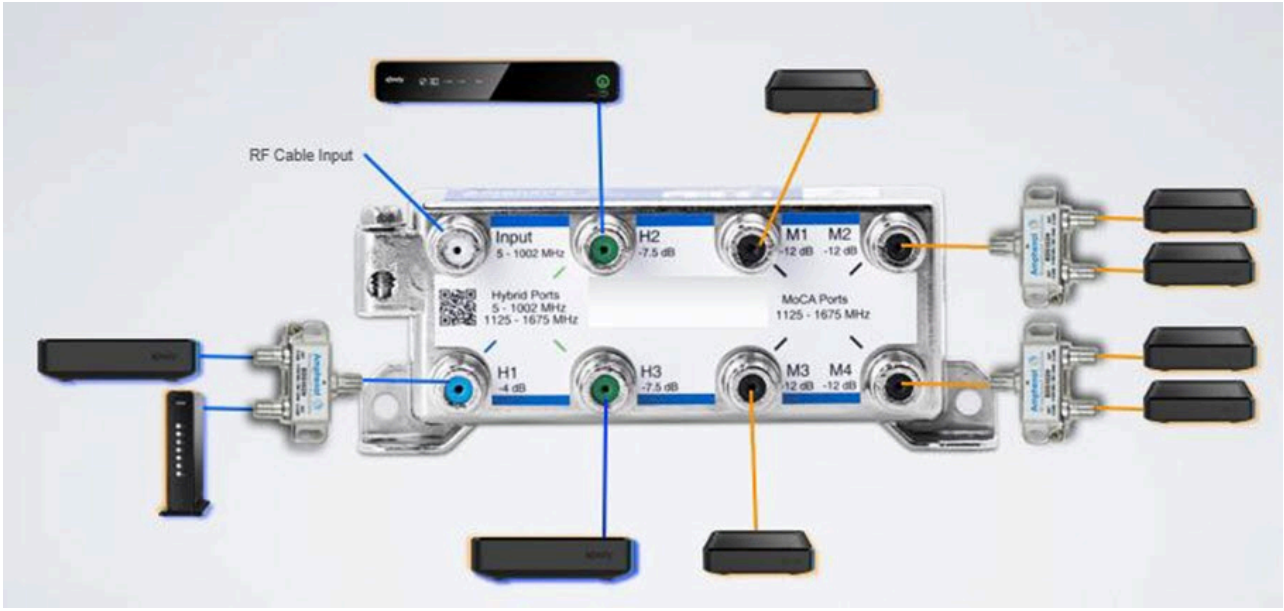
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General outline for signal configuration

- A passive, unamplified Radio Frequency (RF) splitting distribution network shall always be the installation of choice to feed all outlets from the primary service distribution point in the premise.
    - The objective for all RF splitting distribution networks should be to maintain a balanced loss to all outlets when possible.

  - This would include using a Multimedia over Coax Alliance (MoCA) passive device.
- For installations in premises where a single splitter at the primary service distribution is not possible, the cascading of splitters is permitted to a maximum of two on an exception basis.





- RF signal amplification can be used after determining that a passive splitting configuration will not support the minimum specified outlet RF signal levels, and/or the maximum specified upstream transmit levels for each of the active outlets and/or when there are six or more outlets.
    - RF Amplifiers are to be considered as a last resort option only.
    - RF signal amplifiers may only be installed at the primary service distribution point.
    - All RF signal amplifiers must be unity gain, support MoCA and be compliant to the return path being utilized (i.e., mid-split, high-split).
- RF splitting networks shall not utilize directional couplers.
    - Preexisting multi-dwelling unit (MDU) designs will be the only exception to this as defined by Comcast Corporate Engineering.
- Based on the above information, below is the order of preference when completing a distribution network at a customer premise:
    - A balanced passive split network home run to one distribution location
    - An unbalanced passive split network, with a remote split location
    - A balanced MoCA passive split network home run to one distribution location
    - An unbalanced MoCA passive split network, with a remote split location
    - A balanced unity gain amplifier network, amplifier must support MoCA and be compliant to the return path being utilized (i.e., mid-split, high-split, home run to one distribution location)
    - An unbalanced unity gain amplifier network, with a remote split location
- A Point of Entry (POE) RF filter must be installed on every RF distribution network as defined in the MoCA specifications OES.
    - When all IP/FDX technology is utilized for video services, the MOCA filter is to be removed. This is essential for the readiness of advanced technologies such as FDX.
- Unused RF ports in the splitting network at the primary service distribution point and/or outlets must be properly terminated.



  - Only terminators with internal 75-ohm resistors approved by Comcast Corporate Engineering shall be used.
- A permanent, uninterrupted home-run cable connection established between the single, primary service distribution point of the premise and every wired outlet inside the premise shall always be the configuration of choice.
- RF Series-6 cable must be used from the primary service distribution point to every wired RF outlet.

RF signal level ranges and verification methods

- Premise Health Test (PHT) is the minimum required test on all customer visits to validate signal levels and the operation of all Customer Premise Equipment (CPE).
- An RF signal meter (i.e., XM Meter) must be used to determine that all levels in the operational RF spectrum meet the signal level requirements as determined by PHT.
- An egress measuring device, (i.e., Howler), may be required to determine and correct any egress as defined in the Howler standard.
- RF ingress levels must be measured and maintained at or below division engineering levels on the drop network from the tap to the customer outlet(s).
- Signal leakage levels must be maintained at or below 10 µv/m at three meters or its equivalent, and the test frequency must be within the VHF aeronautical band of 108-137 MHz or a frequency in which the results can be correlated to that band.

- If signal levels and/or quality at the tap are going to cause you to configure the premise outside of Comcast standards or are degraded, causing a noticeable impact to the customer's services. Then, follow the guidelines provided in the RTM auto-flow.

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